Wireless Networks
https://doi.org/10.1007/s11276-023-03315-7

ORIGINAL PAPER

IoT-based smart agriculture: an exhaustive study

Avinash Pawar[1] · S. B. Deosarkar[2]

Accepted: 7 March 2023
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2023

Abstract
Despite what some may believe, the agriculture industry is really more accurate, data-driven, and intelligent than ever before in today's modern farming business. In practically each business, particularly "smart agriculture," the fast development of Internet-of-Things (IoT)-based technology has led to a redesign that has switched from statistical to numerical techniques. Such radical developments are upending traditional agricultural practises and opening up new possibilities alongside a number of difficulties. This article discusses the promise of wireless sensor nodes and the Internet of Things (IoT) in agricultural in addition to the difficulties that will inevitably arise when combining this technology with conventional agricultural methods. In-depth analysis is done on IoT devices and communications methods related to wireless sensors used in agricultural applications. There is a list of the sensors that are suitable for various agricultural applications, such as land preparation, crop condition, watering, and pest and insect detection. It is stated how well this technology assists farmers in all crop-related processes, including planting, growing, harvesting, packing, and transportation. This article also takes into account the usage of unmanned drones for agricultural monitoring as well as other advantageous purposes like increasing crop output. Modern IoT-based technologies and architectures utilised in farming also are emphasised where appropriate. We conclude by highlighting future research difficulties and identifying existing and future IoT in agricultural developments based on our in-depth assessment.

Keywords Internet of Things · Climate change · Sensors · Devices · Farming technologies · Unmanned autonomous vehicle · Smart agriculture

1 Introduction

"IoT" is often used to refer to the IoT. The word refers to the routine of utilising the internet to manage and keep tabs on common household gadgets. The term "things" is used to describe any everyday object that may be wired into a network or accessible through the World Wide Web. The Internet of Things (IoT) has the potential to provide end-to-end solutions for any product or service via the use of artificial intelligence (AI), sensors, networking, power, the cloud, and message-passing protocols [1]. According to projections, the world's population would have swelled to about 10 billion by the year 2050. The only way to sustainably feed such a massive population will require agriculture to completely embrace technological advancements. One other critical industry for IoT is agriculture. Monitoring crops and soil conditions is an important part of agriculture, and IoT systems are the best option for doing so. As a consequence of IoT, smart farming has emerged. Utilizing IoT might help farmers cut down on waste while increasing yield. Due to advancements in technology, sensors may now be used to keep tabs on agricultural operations. Farmers could monitor the weather and other conditions in the area [2].

Given the worldwide trend toward new technologies and implementations, it is crucial to achieve the goal of developing agriculture. The agriculture sector has been the focus of several types of research. Typically, this is the practise of using a wireless device network to collect data from a wide variety of sensors deployed over a set of nodes

✉ Avinash Pawar
Knpawar1981@gmail.com

S. B. Deosarkar
sbdeosarkar@yahoo.com

[1] Instrumentation Engineering Deparatment, VPM's MPCOE Velneshwar, Velneshwar, India

[2] Electronics and Telecommunication Engineering, D.B.A.T.U.Lonere, Lonere, India

Published online: 07 April 2023

and send that data using a wireless protocol [3]. Several aspects of the environment may be better understood thanks to the data collected. Increasing agricultural productivity requires careful consideration of factors outside the control of farmers. Many additional factors lead to a steeper decline in output. As a result, agricultural automation is necessary to address these challenges. To address any of these issues, it is necessary to create a holistic system that takes into account all factors that affect productivity at each stage. However, many obstacles prohibit the complete automation of farming. Despite being necessary at the analytical level, it is not made available to farmers as a commodity to maximise resource use. Therefore, this research explores the growth of intelligent agriculture via the Internet of Things and its distribution to farmers.

Until recently, farmers had to rely only on manual methods for managing their farms: measuring soil nutrients, delivering water to crops, inspecting crop quality, and so on. The widespread use of smartphones across India is nothing short of a revolution. The modern-day farmer uses his mobile device for more than just making phone calls; he also uses it to keep tabs on his field. The term "smart agriculture" refers to the practise of using wireless technologies and sensors to control and monitor farming activities [4]. Using sensors and wireless technology, a farmer might get up-to-the-minute data from the field at his off-site position. The integration of wireless sensor networks and environmental modelling allows for more accurate estimates and use of irrigation water in nursery and greenhouse operations, as well as monitoring green roofs for stormwater mitigation [5]. Modern agriculture cannot function without the innovative application of technology known as "smart farming." To realise his dreams of prosperity in life, health, and farming, a farmer must be steadfast in his adherence to a few core principles of sensible farming. By 2040, the world's population is expected to reach around 9.5 billion. Therefore, the agricultural business has to embrace IoT to deal with issues like harsh weather, rising temperatures, and the environmental implications of intensive farming operations if it wants to succeed in such a competitive market. There needs to be enough food to meet the need. However, the agriculture sector benefited from the IoT systems. Several different types of agricultural IT applications and their respective benefits have been addressed. To sum up, IoT will help resolve major economic and environmental concerns while also providing a wide range of employment opportunities. Internet-of-things (IoT) gadgets analyse collected data and then send out the results to customers promptly [6]. Having a firm grasp of how the Internet of Things (IoT) gadgets function in the modern world can simplify tasks, cut down on wasted time and resources, and boost agricultural earnings. In an automated system, machines do the expected tasks while the updated data is saved in the cloud. That is to say, the data may be evaluated remotely from any location where an internet connection and a smart device are available. Instead of paying employees hourly as was common in the past, modern farming pays them a flat rate at the end of the season.It helps in problem-solving and making good decisions. When living in the contemporary technological age, having a robust and trustworthy system eliminates stress and frees up time that can be put to better use. This is the digital age. This means the agricultural sector is likewise in need of digitalization. Hosted cloud services for the Internet of Things (IoT) have become more popular. This paper will cover the methods we're using to develop Internet of Things (IoT) services for the agricultural industry. The major objective is to aggregate data from several nodes so that farmers may wirelessly manage a wide range of farm operations and have access to a respectable, intelligent, and savvy agricultural landscape. Smart agriculture is a relatively new concept, based on the notion that Internet of Things (IoT) sensors might provide data about agricultural regions and take action based on that data in response to human input. The primary contribution of this study is the development, using sensors and an Arduino UNO board, of a system capable of monitoring in-field temperature, water, and moisture levels and even moving if something happens that threatens to harm crops. Smart agriculture is a relatively new concept, including the use of Internet of Things (IoT) sensors to provide data on agricultural regions and take action based on that data in response to user input. This programme is an effort to integrate automated and developing technologies like IoT into farming to create "smart" agriculture. Even after being developed, hardware will need to be revised to accommodate new standards and innovations in the industry. Any updated hardware will be reflected in the latest software release. It's important to put this new version through its paces to make sure that the changes made to the prior version don't cause any problems with other parts of the program. This is crucial since updating a single component might have unintended consequences for the rest of the hardware.

Agricultural sensor utilisation is not a recent phenomenon. As a rule, the sensors used by farmers had one major drawback: They couldn't provide real-time data. The data gathered by the sensors would be stored in a connected memory. Nonetheless, IoT innovation in agriculture resulted in the creation of smart sensors that communicate with the cloud through cellular and satellite networks. This allowed for information to be gathered in real-time, which improved the effectiveness of subsequent decisions. Farmers have increased the effectiveness of their irrigation system by constantly monitoring the water levels in the

tanks. Sensors are being used in almost every step of the agricultural production process, thanks to the proliferation of smart sensing and sensing technologies, connection technologies, and other IoT technologies in recent years. As a result of the IoT, farmers are better able to make choices based on up-to-date information, save money, and make the most of their resources.

The latest estimates put the global population at 9.6 billion by the year 2050. Therefore, farmers are under pressure to produce more food by using resource optimization strategies, despite climate change, harsh weather, etc., and a perpetually rising demand for food.

Because of falling water tables, conserving water is more important than ever. Using data collected from Internet-enabled sensors, farmers might be notified when their crops' water requirements have been satisfied. Power use during peak hours might be lowered with the aid of sensors.

2 The smart agriculture includes the features like:

- Crop health surveillance
- Check the soil's health,
- The quantity of water in the field, and the nutrition in the soil for crop development.
- Controlling the emission of greenhouse gases, etc.
- Proper treatment on climate changes & genetic disorders in seeds

2.1 Few of the challenges in the agricultural industry in India:

- Lack of labour force and resources
- Climate changes & genetic disorders in seeds
- Large-scale manual intervention is necessary
- Inadequate monitoring
- Problems with large-scale unstructured data analysis and inadequate production data.
- ICT (Information and Communication Technology) illiteracy and inadequate ICT infrastructure.
- Drastic changes in the weather
- Among young, educated professionals, there is a lack of interest in the agricultural profession.
- High-priced tools for the job.
- Extra manual labor
- Manually keeping track of records.

This article is a compendium of knowledge that can help the researchers and agriculture engineers implementing the IoT-based technologies to achieve the desired smart agriculture. The rest of this document is organized as follows. Section 2 provides a deep overview of IoT in agriculture for smart farming applications. Section 3 discussed the IoT applications for smart agriculture and farming. Section 4 describes the life cycle of IoT based smart agriculture applications. Section 5 provides the benefits of IoT in agriculture applications. Section 6 discusses the smart agriculture needs and importance. Section 7 discusses the technologies used in smart agriculture. Section 8 illustrates the best practice adopted for IoT agriculture. Section 9 illustrates the IoT based smart agriculture layers and Sect. 10 describes the conclusion and further extension of the paper.

3 Literature review

Farooq et al. (2019) This study describes about the new water crisis scenario, diminishing river and tank supplies, environmental uncertainty, and the urgent need for efficient water management. Sensors are placed at strategic points for crop monitoring to enable this adaptation of temperature and moisture. Researchers reported daily declines infollowing are some of the phases agricultural productivity after conducting field testing [7].

Patil et al. (2017) This paper highlights the use of technology in agriculture significantly impacted both the reduction in labor costs and the increase in productivity. Many separate attempts at analysis impede efforts to help farmers thrive by giving them access to tools that might otherwise boost agricultural output. Using the whole toolkit of computer processing methods made accessible by cloud computing infrastructure, data gathered from farms may be safely transferred to a central database. By connecting a "good sensible intelligent" sensing system with a "smart irrigation system" through wireless communication technology, this idea provides a unique approach to efficient farming. It suggests using a network of inexpensive wireless detectors to monitor soil moisture, humidity, and temperature across many fields and trigger an irrigation system only when the crop needs water [8].

Wang and Chang (2016) In this paper the author proposes a theory explaining the development of automated irrigation systems to optimize water consumption in agriculture. Agriculture is a key economic driver and primary source of income for the vast majority of people in many countries. It is still necessary, however, to compel full technological development and widespread use within the agricultural sector in the Republic of India. Different governments have coordinated efforts to provide farmers with online and mobile electronic communication services for agricultural questions, agro-seller data to farmers, and static information on soil quality in every place. Real-time data on soil quality supported by its current characteristics

may be used in a future system that has not yet been implemented. The soil's quality may be seen in its physical properties. Soil health can be maintained with precise fertilizer application thanks to real-time monitoring [9].

Suma et al. (2017) The author focuses on the soil wet analysis may help with both water availability in times of need and water conservation by identifying when water is still present in the soil. Both agricultural yield and crop diseases are affected by climatic conditions including temperature and precipitation. We need a live model that can collect this kind of information in real time. To increase output and streamline the distribution of agricultural commodities, it is necessary to connect all agricultural institutions, including farmers, advertising agencies, vendors, and consumers. An accountability mechanism like this one will be required to track alternative indicators like agricultural product prices [10].

4 IoT applications in smart agriculture and farming

Here are some of the more compelling examples of how the Internet of Things is already being put to use in farming.

4.1 Real-time crop monitoring

Farmers can monitor crop quality and allocate resources more efficiently with the use of real-time agricultural data made available by smart sensors, motion detectors, motion-sensing cameras, and light detectors.

4.2 IoT analytics in agriculture

Data collected by smart sensors might be put to use in a robotic decision-making and forecasting process. It helps farmers by making decisions in an intelligent, automated manner as opposed to utilizing manual methods or outdated rule-based systems. Farming communities may benefit from predictive analysis and machine learning in two ways: in anticipating and recovering from natural disasters like floods and droughts. Measuring soil quality, temperature, and humidity are just a few examples of the kinds of things that may be analyzed by the Internet of Things.

4.3 Livestock management

In the context of commerce Human labor is essential in the agricultural sector. Since the global economy is constantly shifting, the supply–demand gap has widened. Livestock is an area that must be closely watched. With the use of IoT-based smart tracking, farmers may have stock information sent straight to their smartphones. It aids farmers in identifying flu epidemics in their flocks much more quickly and telling affected breeds apart from healthy ones.

4.4 Drone-based use cases

Commercial drones have several potential applications in precision agriculture. In addition to monitoring the quality of the air, soil, and moisture, the drones are used for physical duties like the automated application of fertilizer, the prevention of physical epidemics in farms, and so on. While there are some challenges in using drones on farms, there is a chance that efficient deployment and use cases may significantly reduce the demand for human labor in the agricultural sector.

4.5 Smart greenhouse solutions

To keep a greenhouse at the perfect temperature and humidity level, constant attention, and manual adjustment are necessary. This process may be automated with the help of IoT solutions designed for the manufacturing sector. Data from smart sensors may be automatically reviewed, and it's possible to implement fully automated decision-making and climate control using systems based on deep learning. These smart sensors may be monitored by SMS or Wi-Fi to keep tabs on environmental conditions and water use [11].

4.6 Smart waste management

One of the main contrast between agriculture and other businesses is the value of industrial waste. To make fertilizers, biological waste from farms may be recycled, and IoT technologies can help with remote, intelligent control of this process. In agriculture, smart sensors might be used to monitor the whole process, and in waste management, they could identify toxic chemicals.

4.7 Agricultural sensors

Modern farmers use sensors, mobile phones, and drones to gather in-depth information about the local environment and available resources. Smart farming sensors placed all over a field collect data on a broad range of factors in the surrounding environment, which is then sent to a central server. This provides very exact data on soil quality, temperature, and other parameters.

4.8 Monitoring climate conditions

This facilitates improved environmental and weather forecasting. Cell phones make it simple for farmers to monitor anything from crops and output to animals and

machines. Smart analysis of massive data sets may also allow for more exact data on cow health, nutrition, and output. Being resistant to the effects of climate change is another trait much sought after by modern farmers. Due to the increasing population and more unpredictable weather, vertical farming techniques are becoming increasingly important. When combined with information from smart IoT devices, this is achievable even in cramped quarters.

4.9 Crop management

Data on crop growth, anomalies, crop health, precipitation levels, humidity, and climatic variables may help in crop rotation, management, and monitoring. There may be a shift in the status quo if farmers are given access to crop management technologies that allow them to make more profitable choices.

4.10 Water management

Crops may be harmed by either too much or too little watering. Irrigation and other methods of water management are thus essential. Connected sensors may aid in decision-making depending on the conditions of the crops, the weather, and the soil moisture. Farmers may then utilize this data to make educated decisions.

4.11 Cattle monitoring and management

Animals may soon have sensors connected to them as part of the IoT to track their heart rates, temperatures, where they graze, how active they are, and what they eat.

4.12 Soil temperature

Temperature in the soil affects the root's capacity for growth, respiration, breakdown, and nitrogen mineralization. Farmers need to employ optimal temperatures for planting and growing their crops to ensure the highest possible output. In the right locations, buried temperature sensors coupled with wireless technologies like infrared may provide farmers with real-time information on soil temperature [12].

4.13 Soil moisture

Water is required for the process of photosynthesis, which all plants must undergo. Soil moisture has a crucial function in regulating soil temperature, which in turn affects crop growth and yield. Subterranean electrode probes provide precise soil moisture monitoring. Internet of Things devices that monitor soil moisture may record and schedule irrigation, as well as gather data on the soil's relative humidity and temperature [13].

4.14 NPK measurements

Nitrogen, phosphate, and potassium levels in the soil may also be measured using Internet of Things sensors. As a result, farmers may determine whether they need to supplement the soil with additional nutrients to maximize crop yields. These sensors, which have just recently become commercially available, use a variety of methods for better precision. The NPK content and soil pH may both be determined with the use of these sensors. These sensors' ability to monitor dynamic soil nutrient changes in real-time makes them valuable tools in agriculture, greenhouses, and soil science.

4.15 Solar radiation

Another use of IoT in agriculture is the monitoring of different types of solar radiation. Solar radiation refers to the amount of energy emitted by the sun that reaches Earth. Because photosynthesis requires it, agricultural productivity will suffer greatly if it is not there. The quantity of sunlight that the soil receives and absorbs influences the rate of evaporation of moisture and the range of temperatures it experiences. Sensors connected to the Internet of Things (IoT) might be used by farmers to track variations in factors including shortwave, ultraviolet (UV), and photosynthetically active radiation (PAR).

4.16 Precision agriculture

When it comes to applying fertilizer, irrigation, and pesticides, precision agriculture is a management technique that leverages digital technology to help farmers with these decisions. Farmers can make more informed decisions about the management of their crops when they have real-time access to data from sensors that track things like temperature, humidity, soil moisture, crop health, and the specific location of assets (RTL).

4.17 Smart pest

Insecticides are released in response to pest detection by control sensors. This may be used with smart irrigation management to save money on pesticides by only spraying the areas that need them.

4.18 Fertilizer management

When sensors detect low fertilizer levels, farmers can consult a crop yield map to determine which areas might

benefit from more fertilizer. As the season progresses, they may keep track of how much fertilizer has been spread over each farm or plot. This reduces costs and lessens pollution from runoff [14].

5 Life Cycle of An IOT Based Smart Agriculture

The following are some of the phases that A Sample IoT Analytics Use Case may go through in the course of its life cycle:

5.1 Selection of sensors

The sensors required for monitoring cattle, for instance, are quite different from the sensors required for a smart greenhouse.

5.2 Data collection

Collecting information from sensors and reformatting it for usage.

5.3 Making decisions and deployment

It's possible to use the information gleaned from the sensors to automate business decisions and give valuable insights. After enough data science is finished, life cycle models are deployed, if required, to the cloud or local servers.

5.4 Recalibration of models

The previous procedures' outcomes are tracked and recalibrated based on the business's key performance indicator and deviation from the result (Fig. 1).

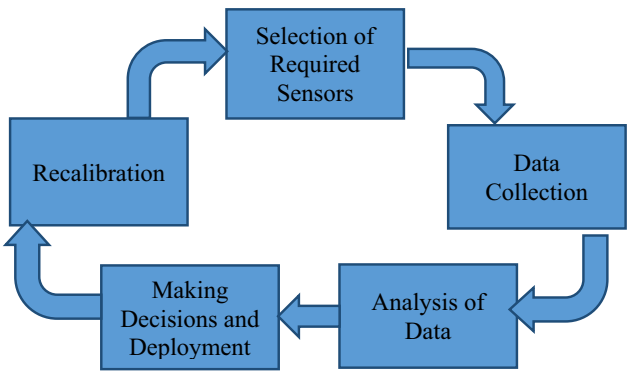


Fig. 1 Life cycle of IOT-based smart agriculture

6 Benefits of IoT in agriculture

To operate in today's environment, technological advancements are necessary. Dependence is projected to increase in the agricultural industry during the next several years. More and more farmers are using IoT-enabled smart technology to boost productivity and efficiency (IoT). Below are just a few advantages brought about by IoT deployments in the agricultural sector:

6.1 Data

In agriculture, one of the key benefits of IoT is the ability to gather data. Farmers can check on the state of their crops and the moisture content of the soil due to sensors placed strategically throughout the farm. This information might be useful for deciding how much water, fertilizer, and pesticide to use.

6.2 Improved quality

Thanks to all of this data, farmers can keep tabs on their crops at all times. That way, they can catch problems in the early stages and take action to resolve them before they become worse. The result is stronger plants that provide better-quality produce.

6.3 Risk reduction

Internet of Things devices may also assist farmers to reduce risks. Farmers, for example, might benefit from monitoring weather patterns to determine optimal planting and harvesting times. This lowers the odds that bad weather may ruin your harvest.

6.4 Business automation

Thanks to IoT, many mundane agricultural operations may be computerized and optimized. For farmers, the benefits in terms of both time and money saved are substantial. To account for variables like as soil moisture, certain automated irrigation systems may modify watering cycles automatically. By doing so, you can be sure that your plants are receiving just the right quantity of water.

6.5 Remote monitoring

Farmers may now keep tabs on their crops' progress from their desktop computers or even their phones. There is a possibility that smart farming systems may send alerts to farmers' mobile devices if there are problems with the soil, seeds, or other parts of their crop [15].

6.6 Drought monitoring

Many farmers face considerable challenges when confronted with drought. Using IoT technology, farmers may foresee water shortages and take preventative measures. Some systems may even recommend the best times and places to water crops.

6.7 Harvesting automation

Robots are increasingly being used for harvesting purposes in the agriculture sector. Therefore, labor costs are decreased while the output quality is enhanced [11].

6.8 Agility

IoT's use in farming helps boost operational agility, which is a huge boon to the industry. Farmers can react promptly to changes in the weather, humidity, air quality, and the condition of each crop and soil thanks to real-time monitoring and forecasting systems. Professional farmers have access to more resources than ever before to safeguard their crops from hazards like harsh weather.

7 Need for smart agriculture

The current global population of little over seven billion will have increased to nine or ten billion by 2050. There won't be enough food for everyone if we don't use sustainable farming practices immediately. Internet of Things (IoT) sensors may be used to keep tabs on crops so that farmers always know when to water and fertilize plants with just the right amount. This helps the environment by reducing waste and increasing food production via environmentally friendly methods that benefit humans and animals alike.

7.1 Understanding predictive analytics for smart farming

Predictive analytics, which the Internet of Things may give, is essential for comprehending the food production process. Farmers use predictive analytics to track crop growth over time, allowing them to make informed decisions about whether or not to begin a new planting cycle and whether or not their current crops will be adequately watered. Predictive analytics also helps farmers learn which seeds thrive in a range of soil and weather conditions, as well as in different places. Some could claim that smart sensors would keep us from starving and guarantee sustainability at every turn [16].

7.2 Data security in the agriculture industry

The significance of data security in the agricultural business cannot be overstated. Information gathered from a single farm may be private due to the nature of the data itself and the details it contains, such as the crops cultivated, the amount of water and fertilizer used, and other details specific to that farm. The information must be kept confidential to prevent farmers from poaching one other's customers and proprietary information. To gather these crucial agricultural data, the Internet of Things (IoT) devices must be equipped with strong cybersecurity safeguards, such as encryption and password protection features [17].

7.3 Using end-to-end farm management system

Farm management systems are essential to the success of any modern agricultural business. They are created to help farmers decide when to harvest and when to apply pesticides to their crops. In addition, they allow them to keep tabs on and analyze a plethora of data emanating from the Internet of Things sensors installed on their premises. The question is, how exactly do these farm management systems function? A farmer needs an end-to-end system, which requires the installation of communication infrastructure like GSM/GPRS cellular networks in his fields (or the use of satellite communication). A gateway device with a strong wireless connection is then set up with the cloud-based software. A farmer may then use this technique to maintain tabs on his crops, predict when they'll be ready to harvest, and control water use. Farmers may also use remote-control technology to run harvesting tractors and milk vans that deliver milk from faraway farms.

8 Technologies used in IoT-enabled smart agriculture

The following components make up a typical Internet of Things system used in farming:

- Sensors
- Signal conditioning
- Processing & Security
- Power Management
- Connectivity
- Positioning

8.1 Sensors

Sensing devices are the lifeblood of every IoT system since their data is the system's most vital asset.

Accelerometers can track motion, detect encroachment into farmland, and measure activity.

Soil, air, and animal health may all be monitored with the use of humidity and temperature sensors. These sensors provide very useful data for allowing IoT applications in agriculture.

Due to its sensitive capacity to measure atmospheric pressure, a pressure sensor may also be used to make rough height calculations. Since a diaphragm must fit through an opening in a pressure sensor, the device is vulnerable to chemical and water damage. Pressure sensors must be resistant to common pool chemicals like chlorine and bromine [18].

8.2 Connectivity

Consider the following list of uses for networked devices in agriculture.

8.2.1 Range

Short-range radio frequency (RF) technologies like Bluetooth low energy (BLE) and Wi-Fi can gather data from sensor nodes and send it to a gateway, making them ideal for use in interior monitoring. Long-range options like GPRS and LPWAN (like Sigfox) are used for monitoring big farms because of their proximity to existing telecommunications infrastructure. Short-range technologies may be used in the great outdoors if their range is extended using mesh networking protocols like BLE Mesh and 6LowPan.

8.2.2 Power consumption

For this reason, Sigfox and other LPWAN technologies are favored over Wi-Fi, Bluetooth, and GPRS since most agricultural applications rely on non-rechargeable batteries. Therefore, in our research, we zero down on BLE and sub-GHz since they seem to be the most relevant communication technologies for the agricultural context.

8.2.3 Bluetooth low energy [BLE]

Technology allows smartphones to connect to items while using far less power than Wi-Fi and Bluetooth do. The 2.4 GHz ISM band it uses is freely accessible in most parts of the world. To connect a large number of BLE devices, BLE Mesh was developed as a software solution for IoT mesh networks. Mesh networking allows for connections even when no node in the network has a direct line of sight to any other node. The BlueNRG-2 from STMicroelectronics is a Bluetooth Low Energy (BLE) 5.0 certified ultra-low power system-on-chip that allows for dependable BLE connections between IoT devices, as well as seamless connections to sensors, privacy 1.2, secure connection 4.2, 8 dBm of output power, and more. Furthermore, SIG BLE Mesh is supported in its entirety [19].

8.2.4 Low-power wide area network (LPWAN)

These innovations mark a sea change in communication since they enable links to be established across vast distances (thousands of kilometrers) while using so little power that a common, non-rechargeable coin cell battery may keep them operating for years. Sigfox is a kind of low-power wide-area network (LPWAN) that permits data storage and transmission in the cloud across great distances and international boundaries with little power consumption. The S2-LP solution from STMicroelectronics is an innovative hybrid of Bluetooth Low Energy (BLE) and Sigfox radio, combining the best aspects of both technologies to benefit end users. The dual-band radio supports both the short-range Bluetooth Low Energy (BLE) connection with a smartphone and the long-range Sigfox connection. When a smartphone is paired with a Sigfox device, it is possible to do over-the-air software updates, direct setup, and control.

8.3 Robotics

It's human nature to rely on machines for everything. The capacity to do complex tasks and increase output, on the other hand, only improves over time. Due to escalating demands and global personnel restrictions, farmers have urged agricultural robots to aid in crop production. The falling rates of agricultural production leave farmers no choice but to adopt these agri robot systems. The recent advancement of sensor technology and artificial intelligence that allows machine learning has given agri robots a greater role to play in the process of raising food production. Now that we have some of these things to test, we can learn more about them and dig deeper into their mysteries.

8.4 Weeding robots

Intelligent agri robots called "weeding robots" can identify similarities between crops and weeds in photos, and then use their robotic arms to spray the plants with herbicide. The uptick in plant pesticide resistance is a useful technology for farmers and the environment [20].

8.5 Machine navigation

A farmer may be able to operate his field tractors and heavy plowing equipment from the convenience of his house, much as he could a toy vehicle, using a global positioning system (GPS). This intelligently integrated auto equipment can self-adjust to the different terrains they detect, enhancing precision and lowering the need for human work. Using a smartphone, it is simple to track the location of moving gear, identify and eliminate obstructions, and assess production levels.

8.6 Harvesting robotics

As a solution to the problem of a lack of available labor, ag robots are utilized for crop harvesting and other tasks related to agriculture. Careful harvesting of fruits and vegetables is no problem for these automated systems, which may operate around the clock, every day of the week. These harvesting robots use a combination of robotic arms and image processing to reliably and accurately identify ripe fruits for picking. High-value greenhouse crops like strawberries, apples, and tomatoes might benefit from these robots' ability to accurately predict when to harvest.

8.7 Material handling

The Internet of Things (IoT) enables smart agricultural monitoring systems that effectively manage human labor. They are in good enough health to join the rest of the workforce out in the field. They can improve plant quality, land area, and plant spacing at exact lengths, and they can carry heavy things with ease. With maybe just one change, production costs might be drastically lowered.

8.8 Drones

Currently, drones have the potential to be useful in many different fields. In the field, drones equipped with sensors and cameras are used for a variety of purposes, including land surveys, crop mapping, and aerial photography. Drones are used both on the ground and in the air in smart IoT farms. Drones that operate on the ground are mobile robots with cameras and wheels. Unmanned aerial vehicles (UAVs) or unmanned aircraft systems (UASs) are robot aircraft that are used to read, map, photograph, and navigate a field to gather agricultural production data. Both drones are equipped with technology that allows them to navigate their surroundings autonomously or with human guidance, using either manual controls or those generated by the drone's companion software. There are built-in sensors and a GPS in them. As part of a regular field survey service, they might be charged and maintained at a location close to the farms. Date-specific information on crop status, irrigation, spraying, planting, soil and field, plant count, expected yield, etc. is provided [21].

8.9 Remote sensing

In agriculture, the Internet of Things may be used for things like remote sensing, which gathers data from sensors placed all across fields. The information is then sent to analytical tools for further examination. Sensors are high-sensitivity devices used to detect deviations from the norm. To keep a watch on their crops and respond appropriately based on analytical dashboard data, farmers use sensors and monitors in equipment built for smart IoT farming. Various sensors monitor the environment, the plants, and the ground. It provides extensive data to farmers on the condition of their farms and fields. The farmers are instantly notified of any abnormalities in light, humidity, temperature, crop shape, crop size, etc. The use of remote sensing technology would allow us to keep tabs on agricultural growth and production as well as detect and contain disease outbreaks. Soil analysis also reveals the arid regions and nutrient-rich soil composition of the farm. Knowing the soil's acidity or drainage capability may help the farmer save a tonne of money by tailoring the irrigation system to the area.

8.10 Computer imaging

Computerized imaging systems in agriculture make use of sensor cameras. These are installed on camera-equipped drones or at other field sites, allowing for the automated creation of photographs. These image-processing applications showcase imagery captured at very high intensities throughout the spectrum, including infrared. Through the use of images stored in a database, image processing in tandem with machine learning analyses and organizes crops based on differences in their dimensions, outlines, colors, and rates of growth. This aids in quality control monitoring and keeps farmers' output steady.

9 Best practice to adopt IoT in agriculture

Some examples of how farmers might leverage IoT best practices to advance their enterprises are:

9.1 Choosing the right hardware

Choosing the sensors, devices, and databases (whether on-premises or in the cloud) that are most suited to your

solution is essential for building a solid Internet of Things (IoT) ecosystem.

9.2 Connectivity

To guarantee continuity, reliable communication methods between nodes in the chain are necessary even if all the aforementioned conditions are met.

9.3 Leveraging analytics

The information gathered is useless if the recommended course of action is not followed. Acquiring data analytics tools at the same time as your IoT devices is vital if you want to make sense of the data received and effectively build upon the information gained. Analytics allow farmers to anticipate yield and quality, as well as assess how resilient their crops are to climatic shocks. Using this information, the farmer may make a well-informed decision.

9.4 Monitoring IoT architecture

If you're going to rely on technology to take care of boring tasks like watering, planting, and harvesting so you can concentrate on more important concerns, you need to invest in suitable IoT monitoring solutions that keep tabs on the devices' functioning, continuity, health, and progress around the clock.

9.5 Ensuring data security IoT technology

Works with large data sets from all areas of your business, regardless of your field. Your network security is at risk, and your information might be stolen. One scenario is a drone transferring data to farm machinery over the internet with almost no safeguards in place. To counteract these threats, your digital business should seriously consider investing in cybersecurity technology that safeguards its data and systems [22].

10 IoT based smart agriculture layers

The following articles discuss the advantages and disadvantages of using the Internet of Things (IoT) in farming, as well as the difficulties of using automated machinery in place of human labor. They are sorted and analyzed using a categorization system with four distinct groups. An application's layers, a middleware platform's, a sensor's, and a network is described below (Communication Layer) (Fig. 2).

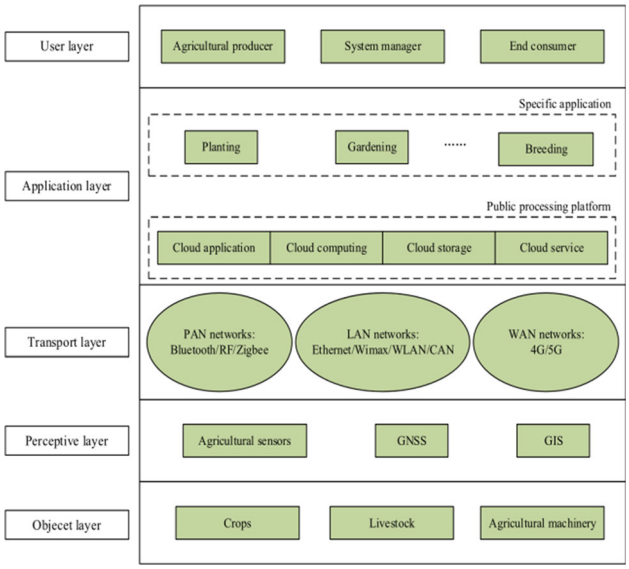


Fig. 2 Layers of IOT-based smart agriculture

10.1 Sensor layer

In agriculture, several processes must be coordinated. Irrigation, pesticides (against insects and fungi), herbicides (against weeds), fungicides (against fungi), insecticides (against insects), soil preparation, harvesting, and storing are all part of the process. Multiple sensors are used to collect information on these operations. These sensors' principal uses in automating the system fall into the following categories.

10.1.1 Leaf sensors

A biometric gadget known as a leaf sensor measures water loss or the stress caused by water shortage in plants. For this purpose, the sensor makes use of the existing moisture content of plant leaves. In particular, this sensor technology is helpful for water conservation since it can detect when plants are beginning to show signs of water stress early on.

10.1.2 Soil moisture sensor

Instruments called "soil moisture sensors" monitor the level of moisture in the ground in real-time. A substantial portion of the irrigation system's ability to allow for precise scheduling of water distribution and delivery is due to the presence of sensors. These signs may be used to adjust watering schedules for optimal plant development [23].

10.1.3 Temperature sensor

The role temperature plays in agriculture. Both very old and extremely warm conditions normally stunt plant

growth. It has a devastating effect on crop production. Agricultural sensors can detect changes in the temperature of the soil, water, and plants. Soil temperatures between 20 and 30 degrees Celsius are optimal for plant growth, whereas water temperatures between 15 and 30 degrees Celsius are optimum [24].

10.1.4 Humidity sensor

Typically, humidity sensors are used to determine how moist the air is. This sensor may also be used to gauge the temperature of the surrounding air. Humidity sensors need two types of parameters to function. Alterations to the flow of electricity follow changes in air temperature.

10.1.5 Light intensity sensors

A common installation for this sensor is in plant factories and greenhouses. Even though all plants need sunlight, the diverse plant groups have evolved unique physiologies to react to and adapt to varying light levels. However, some plants can survive on far less.

10.1.6 Optical sensors/Illumination sensor

Data on the plants and soil may be gathered using these sensors during the whole growing period. Near-infrared, mid-infrared, and polarised light spectra are used by optical sensors to assess soil quality. Sweetness, acidity, firmness, and color may all be evaluated using optical sensors.

10.1.7 Air flow sensors

These sensors are used to measure air permeability in the soil. These air flow sensors have been used in a variety of studies. Soil composition, moisture content, and soil structure/compaction were experimentally distinguished.

10.1.8 Carbon dioxide sensors

Carbn dioxide gas levels in the atmosphere may be tracked with the use of this technique. This sensor is often used in greenhouses and other environments where plants are cultivated. The amount of infrared radiation (IR) absorbed by CO2 molecules may be used as a proxy for the concentration of CO2 in the greenhouse or plant factory.

10.1.9 Soil pH sensor

One of the most important factors in crop yield is the soil's acidity or alkalinity. Crop yield is affected by the soil's physical, chemical, and biological activity and features.

10.1.10 Acoustic sensor

In modern agriculture, acoustic sensors are most often used for monitoring and detecting pests, as well as recognizing seed types based on their unique sound absorption spectra. The tool's contact noise with different surfaces is measured and analyzed to determine how effective the tool is.

10.1.11 Optoelectronic sensor

Sensors like this can track individual plants' development over time. Therefore, this sensor might be useful for farmers in detecting weeds, herbicides, and other unwanted plants in wide-row crops. It can generate a detailed map of weed distribution by combining data from an optoelectronic sensor and the precise location of farms. Furthermore, optoelectronic sensors can differentiate between plants and soils by analyzing their distinctive reflection spectra.

10.1.12 Ultrasonic sensor

Ultrasonic sensors may be used in a variety of settings. These sensors, when combined with a camera, might be used to detect and identify unwanted plants, such as weeds. Ultrasonic sensors are used for level determination, fertilizer application, and crop monitoring.

10.1.13 Field-programmable gate array-based sensor

The sensor is very effective and has widespread use. Despite their high power consumption, these sensors have a wide range of potential uses, such as real-time plant water usage, humidity, soil moisture, and temperature monitoring. They increase costs without improving output.

10.1.14 Eddy covariance-based sensor

Sensors of this kind might monitor energy and gas transfers from the Earth's surface to the atmosphere, including those of methane, carbon dioxide, and water vapor. These sensors are used by most large plantations.

10.1.15 Ion selective electrode sensors

These sensors allow researchers to keep an eye on nitrogen levels in the soil, which is an important ingredient (nitrate, ammonium, potassium, and sodium). The difference in potential between the electrodes is used to quantify this.

10.1.16 Passive infrared sensors(PIR Sensors)

When used in the field, these sensors can tell the difference between humans and other animals.

10.1.16.1 Network Layer(Communication Layer) Communication between layers depends on the network layer. In this layer, only summarised data is delivered to the machine's brain, protecting private information from prying eyes. First and foremost, we want to close the knowledge gap between the software and hardware levels (such as sensors). You may classify modes of information transfer as either one or the other. I'll list a few of them below:u

- Wired Communication.
- Wireless Communication.

There are typically three categories of smart agricultural sensors based on their transmission data rates and power requirements [25].

- Reduced power requirements and data volume In this range, data transfer rates in the hundreds of bytes per second are typical. This layer consumes less than 1 mA of power. Transfer sensors were used to collect data in this area (Leaf sensors, Temperature sensors) [26].
- Medium-Sized, High-Efficiency Data In this class, speeds in the tens of megabits per second are expected. A current of around 10 s mA is used by this layer. In this section, we'll be discussing the methods through which photographs are sent.
- Massive, resource-intensive data
- In this range, tens of megabits per minute of data transfer speed are not out of the question. This layer consumes around 50 s A of current. Streaming video data is sent in this category. To deliver the services necessary for smart agriculture, three interacting levels of end-to-end data transmission solutions may be necessary.
- Layer One Integral to the Smart Agriculture System is the backbone, or "core network," which connects and facilitates a data flow between individual networks. Communication at the Backbone layer is often accomplished using hardwired connections. You could find some of those things at this tier.

  UAV (Drone)
  Base Station

- Intermediary Layer In this level, we use the following methods for data transfer:

- **Zigbee** Throughout the Internet of Things (IoT), Zigbee has emerged as an open, global standard for low-cost and low-power wireless networks (based on IEEE 802.15.4) [27].
- **Bluetooth** Bluetooth is a wireless technology standard that uses UHF radio waves in the ISM bands (2.40 GHz to 2.48 GHz) to set up personal area networks and exchange data over short distances between fixed and mobile devices (PAN).
- **LoRa (Short Range)** As a long-range, low-power wireless technology, LoRa finds widespread use in Internet of Things (IoT) contexts. It employs LPWAN (Low Power Wide Area Network) technology, which requires less power, to link wireless sensors to the cloud.
- **Sigfox** French global network operator Sigfox launched in 2010 to meet the need for reliable network connectivity for low-power devices (or "things"). To encode data, it modifies the phase of the carrier radio wave across small or ultra-thin bands of the electromagnetic spectrum. Zigfox has shown that it can send data in real-time from one hundred sensors.
- **RFID (Radio-frequency identification)** To track the whereabouts and functioning of things, radio-frequency identification (RFID) utilizes electromagnetic waves to scan tags affixed to them. In a radio frequency identification system, the transmitter, the receiver, and the transponder are the three main parts (which control and monitor equipment) [28].
- **4G** Now in existence are wireless networks of the third generation (3G), with speeds around ten times slower than those offered by the fourth generation (4G).
- **WIFI** In its complete form, the abbreviation "WiFi" refers to a technology that enables high-quality wireless connections. The term "wireless fidelity," or simply "Wi-Fi," refers to a kind of networking technology that enables very fast data transfer rates via a local area network. A wide variety of other technologies, including NFC, BLE, 6LoWPAN, Z-Wave, EDGE, ISM/SRD860, Thread, Weightless-N/W, NB-IoT, and LR-WPAN, are used as bridge layers.

10.2 Middleware layer

Connecting the network layer and the application layer is the middleware layer. To simplify the process of integrating new and current systems, middleware is used to mask the intricacies of the underlying operating system, primary network, and sub-smart agricultural system (Sub Sensor System) networks. Several crucial functions, including device management, data storage, analytics, and API access, are all part of this layer (API) In this tier, there are

data centers located in different time zones. Some examples of suitable data storage for IoT projects include Badger, Berkeley DB, LevelDB, ObjectiveBox, Realm, Redis RocksDB, SQLite, EC2, and Azure. Information may also be saved in locally installed databases. Any information sent at the transport layer may be retrieved by using them. Aside from web services and APIs, this layer also contains the underlying infrastructure. These automated processes are outlined in the specifications provided by web services and APIs. Their analysis of a massive dataset may result in competing criteria. If farmers do a good job of analyzing the data, they may be able to more accurately predict their future revenue. The information stored on data servers may be analyzed using techniques like big data analysis [29].

10.3 Application layer

This layer facilitates communication by supplying the required protocols and handling user interactions.To get access to the data given by the middleware, the user must sign into a mobile or web-based application (Android apps and IOS apps). All of these websites and apps supplied encrypted user data, which they decrypted and stored on their servers. Simply said, nodes are the foundation of every wireless network. The following are some working samples of the application layer code. Smart farming, animal management, intelligent lighting, waste management, greenhouse management, and weather monitoring systems are just a few examples of the kinds of technology used in this area [30].

11 Conclusion

To fulfil the rising food need of the expanding global population in the face of the ever-dwindling arable land, the emphasis must be on smarter, faster, and more efficient harvest technologies. The adoption of agriculture as a profession by tech-savvy, creative youngsters, agricultural production as a means of achieving fossil energy individuality, monitoring crop production, safety and nutrient labelling, and collaborations among growers, suppliers, retailers, and buyers are all examples of the creation of new techniques to improve crop yield and ability to handle. In order to improve agriculture's intelligence and efficiency in order to meet the future demands, this article took into account all of these factors and emphasised the importance of numerous technologies, particularly IoT. Wireless sensors, UAVs, cloud computing, and communication technologies are all fully covered for this purpose. A fuller understanding of contemporary research initiatives is also given. Additionally, several IoT-based systems and architecture are offered with regard to farm applications. To aid investigators and engineers, a list of the industry's present problems is included, along with predictions for the future.All of this leads to the conclusion that every square foot of farmland is essential to maximising crop yield. The adoption of environmentally friendly IoT-based communication and sensing technology, therefore, is not an option—it is required to deal through each inch in accordance.

Declarations

Conflict of interest The authors declares that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

References

1. Perera, C., Zaslavsky, A., Christen, P., & Georgakopoulos, D. (2013). "Context aware computing for the Internet of Things: a survey," IEEE Communications Surveys & Tutorials, submitted 2013.
2. Gubbi, J., Buyya, R., Marusic, S., & Palaniswami, M.: Internet of things (IoT): A vision, architectural elements, and future directions, Future Generation Computer Systems, 2013, pp. 1645–1660.
3. Bhatia, S., Patel, S. (2015). "Analysis on different Data mining Techniques and algorithms used in IoT", ISSN: 2248–9622, Vol. 5, Issue 11, (Part - 1) November, pp.82–85.
4. Patil, K.A., Kale, N.R. (2016). A model for smart agriculture using IoT IEEE, December.
5. Shirsath, D. O., Kamble, P., Mane, R., Kolap, A., & More, R. S. (2017). IOT based smart greenhouse automation using arduino. *International Journal of Innovative Research in Computer Science & Technology (IJIRCST)*. https://doi.org/10.21276/ijircst.2017.5.2.4
6. Ya Wang, Y.Y. (2020). Research on Application of Smart Agriculture in Cotton Production Management, In *International workshop on electronic communication and artificial intelligence (IWECAI)*, p. 122.
7. Farooq, M.S., Riaz, S., Abid, A., Abid, K., Naeem, M.A. (2019). A Survey on the role of IoT in agriculture for the implementation of smart farming," Special section on new technologies for smart farming 4.0: Research Challenges And Opportunities, vol. 7, p. 156251.
8. Patil, G. L., Gawande, P. S., & Bag, R. V. (2017). Smart agriculture system based on IoT and its social impact. *International Journal of Computer Applications, 176*, 0975–8887.
9. Wang, Y., Chang, Z., (2016). System of Wireless Temperature and Humidity Monitoring based on Arduino Uno platform, In *Sixth International Conference on Instrumentation & Measurement, Computer, Communication and Control*, pp. 770–771.
10. Suma, N., Samson, S. R., Saranya, S., Shanmugapriya, G., Subhashri R. (2017). IOT based smart agriculture monitoring system IJRITCC, February.
11. Alzubi, J. A., Manikandan, R., Alzubi, O. A., Gayathri, N., & Patan, R. (2019). A survey of specific IoT applications. *International Journal on Emerging Technologies, 10*(1), 47–53.

12. Cuihong Liu W. R. B. Z. L. (2011). "The application of soil temperature measurement by LM35 temperature sensors," In *International Conference on Electronic & Mechanical Engineering and Information Technology*, pp. 1825–1826,.
13. Kumar, M.S., Chandra, T.R., Kumar, D.P., Manikandan, M.S. (2016). "Monitoring moisture of soil using low-cost homemade Soil Moisture Sensor and Arduino UNO," In *3rd International conference on advanced computing and communication systems (ICACCS–2016)*, pp. 1–2.
14. Jain, R., Goel, V., Rekhi, J.K., Alzubi, J.A. (2019). IoT-based green building: towards an energy-efficient future." Green building management and smart automation. IGI Global, 2020. pp. 184–207. Web. 25 June
15. Sathish Kumar, L., Ramanan, M., Jafar, A., Alzubi, P., Jayarajan, Thenmozhi, S. (2021). Smart metering using IoT and ICT for sustainable seller consumer in smart city In *Challenges and Solutions for Sustainable Smart City Development*, Springer, Cham, 2021, pp. 75–89.
16. Ashir, D.M., Talukder, M.T., & Rahman, T. (2022). Internet of Things (IoT) based smart agriculture aiming to achieve sustainable goals. ArXiv, abs/2206.06300.
17. Moghavvemi, M., Seng, L.C.S., Moghavvemi, M. (2004). "Pyroelectric Infrared Sensor For Intruder Detection," In *IEEE Region 10 conference TECON 2004*, pp. 656–657.
18. Dhanaraju, M., Chenniappan, P., Ramalingam, K., Pazhanivelan, S., & Kaliaperumal, R. (2022). Smart farming: Internet of Things (IoT)-based sustainable agriculture. Agriculture.
19. Yenikaya, M. A., Güvenoğlu, E., & Kondakci, S. (2022). Internet of things (IoT) based smart irrigation and fertilization system. *Türkiye Bilişim Vakfı Bilgisayar Bilimleri ve Mühendisliği Dergisi*. https://doi.org/10.54525/tbbmd.1028785
20. Viswanathan, S., & Sathesh Kumar, K. (2022). Role of Internet-of-Things (IoT) and sensor devices in smart agriculture: A survey. In *2022 6th International Conference on Intelligent Computing and Control Systems (ICICCS)*, pp. 421–424.
21. Ma, Z., Rayhana, R., Liu, Z., Xiao, G. G., Ruan, Y., & Sangha, J. S. (2022). Industrial internet of things (IoT) and 3D reconstruction empowered smart agriculture system. *IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS), 2022*, 311–316.
22. Gupta, M., Abdelsalam, M., Khorsandroo, S., Mittal, S. Security and Privacy in Smart Farming Challenges and Opportunities, vol. 8, pp. 34567–34568, 2020.
23. Adamchuk, V. I., Hummel, J. W., Morgan, M. T., et al. (2004). On-the-go soil sensors for precision agriculture. *Computers and Electronics in Agriculture, 44*(1), 71–91. https://doi.org/10.1016/j.compag.2004.03.002
24. Movassagh, A. A., Alzubi, J. A., Gheisari, M., Rahimi, M., Mohan, S., Abbasi, A. A., & Nabipour, N. (2021). Artificial neural networks training algorithm integrating invasive weed optimization with diferential evolutionary model. *Journal of Ambient Intelligence Humanized Computing*. https://doi.org/10.1007/s12652-020-02623-6
25. Gheisari, M., Yaraziz, M.S., Alzubi, J.A., Fernández-Campusano. C., Feylizadeh, M.R., Pirasteh, S., Abbasi, A.A., Liu, Y., Lee, C.C. (2022). An efficient cluster head selection for wireless sensor network-based smart agriculture systems In *Computers and Electronics in Agriculture*, vol 198.
26. Bowman, K. D. (2010). Longevity of radio frequency identification device microchips in citrus trees. *Hortscience, 45*(3), 451–452. https://doi.org/10.21273/HORTSCI.45.3.451
27. Zhang, Q., Sun, Y., Cui, Z. (2010). Application and analysis of zigbee technology for smart grid," In *International Conference on Computer and Information Application(ICCIA 2010)*, pp. 171–174
28. Ampatzidis, Y. G., & Vouqioukas, S. G. (2009). Field experiments for evaluating the incorporation of RFID and barcode registration and digital weighing technologies in manual fruit harvesting. *Computers and Electronics in Agriculture, 66*(2), 166–172. https://doi.org/10.1016/j.compag.2009.01.008
29. Luong, N. C., Hoang, D. T., Wang, P., Niyato, D., Kim, D. I., & Han Z.: Data collection and wireless communication in the internet of things (IoT) using economic analysis and pricing models: A survey arXiv:1608.03475v1 [cs.GT] 11 Aug 2016.
30. Newlin Rajkumar, M., Abinaya, S., & Venkatesa Kumar, V. (2017). Intelligent irrigation system – an IoT based approach IEEE International Conference on Innovations in Green Energy and Healthcare Technologies (ICIGEHT'17), 978–1–5090–5778–8/17/$31.00©2017 IEEE.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.



Avinash Pawar had received his Master Degree, ME from the Savitribai Phule Pune University, Pune, India, in 2019. He is currently working as Assistant Professor at the Department of Instrumentation Engineering, VPM's Maharshi Parshuram College of Engineering, Velneshwar, India affiliated to Mumbai University. At present he is working as a Ph.D Research scholar at Dr. Babasaheb Ambedkar Technological University, Lonere, India. His research interests mainly include Internet of Things, Wireless Sensor Network and Signals and System.



S.B. Deosarkar had received Ph.D. degree from the Swami Ramanand Teerth Marathwada University, Nanded, India, in 2004. He is currently a Full Professor with the Department of E and TC Engineering, Dr. Babasaheb Ambedkar Technological University, Lonere, India. He has to his credits several research papers in various National and international journals of high impact factor and had authored several book chapters in the domain of Communication Engineering and healthcare. His research interests include Microwave Communication, EMI/EMC, Microstrip Antenna Design, Optics and Sensor systems. He had guided more than 12 PhD research scholars and had received funding to his several Research Proposals from various funding agencies.